TinyML and Edge AI for power grid management

"pull up some research papers related to the usage of tiny ml and edge ai technology to improve grid management." *Generated by LeapSpace*

TinyML and Edge AI for Grid Management at a Glance

Key Findings Table

Area	What the evidence shows	Example papers / citations
Edge forecasting	Edge and TinyML methods are being used for load, solar, and inverter forecasting to reduce dependence on remote cloud processing; reported accuracy gains are real, but the best model class varies by dataset and task [1] [2] [3] [4] [5] [6] [7] [8].	Tiny Transformer for GHI forecasting [1]; LightGBM load forecasting [4]; CNN-LSTM [5]
Fault/anomaly detection	TinyML-style local inference is well aligned with anomaly and fault monitoring because it supports low-latency response on constrained devices, with Raspberry Pi and ESP32-class feasibility reported [9] [10] [11] [12] [13].	Raspberry Pi anomaly detection [11]; ESP32 deployment [12]
Predictive maintenance	Energy-asset maintenance is an active use case, especially for PV-related systems, but the literature still has a gap between prototype demonstrations and utility-grade, asset-specific reporting [14] [15] [16].	PV TinyML review [14]; Edge-AI maintenance example [16]
Voltage/frequency regulation and control	Edge AI is most compelling for microgrid control loops, where local controllers reduce latency and several AI-based regulators outperform conventional PID or manually tuned droop approaches [2] [17] [18] [19] [20].	DRL voltage regulator [17]; droop tuning [19]
Architecture choice	Hybrid edge-cloud systems repeatedly emerge as the practical middle ground: keep latency-sensitive inference/control local, use cloud/fog for coordination, redundancy, and model management [21] [22] [23] [24] [25].	Edge-cloud forecasting [22]; fog/cloud EMS [25]

Area	What the evidence shows	Example papers / citations
Hardware constraints	The strongest hardware evidence is mostly generic TinyML rather than grid-specific; memory, latency, and energy trade-offs drive model compression and hardware-aware selection [1] [26] [27] [28] [29] [30] [31].	Pareto benchmarking [29] [31]
Federated learning pipelines	Federated approaches are being tested for anomaly detection, PV prediction, and microgrid energy management to preserve privacy and reduce data movement, but non-IID data, link unreliability, and adversarial threats remain central concerns [32] [33] [34] [35] [36] [37] [38].	FL anomaly detection [32]; hierarchical FL [39] [40]

Direct Answer

Yes—there is a solid body of research relevant to using TinyML and edge AI to improve grid management, especially in **three practical buckets**: **forecasting**, **fault/anomaly detection**, and **latency-sensitive microgrid control** [1] [2] [9]. The strongest papers to pull up from the supplied evidence are:

- **Forecasting and local grid intelligence**
  - Tiny Transformer for on-site solar irradiance/GHI forecasting, reported as better than RNN/ANN baselines on model size, inference time, and accuracy [1].
  - ARM-based smart-meter-concentrator deployment for microgrid solar inverter forecasting plus droop control [2].
  - Comparative short-term load forecasting papers showing mixed winners across **LightGBM, XGBoost, LSTM, CNN-LSTM, and GRU** rather than a single dominant architecture [4] [5] [6] [7] [8].
- **Fault detection / predictive maintenance**
  - Smart-grid and PV-focused surveys positioning TinyML and edge AI as good fits for local anomaly detection and maintenance [9] [10] [14] [15].
  - Hardware-constrained deployments with reported latency/memory feasibility on **Raspberry Pi** and **ESP32-class** devices [11] [12] [13].
- **Microgrid control**
  - DRL- and ML-based controllers for inverter voltage and droop regulation that outperform PID or conventional tuning on deviation, oscillation, or settling behavior [17] [18] [19] [20].
  - Comparative control-architecture studies showing lower latency, less bandwidth use, and better resilience for edge or hybrid designs than cloud-only control [25] [41] [42] [43].

Taken together, these papers suggest that **the best near-term grid use of TinyML/edge AI is not "do everything on-device," but "keep fast decisions local and use cloud/federated layers for retraining, coordination, and security management"** [22] [24] [39].
Confidence level: High - supported by convergent findings across forecasting, control, and deployment architecture studies [2] [10] [25].

Study Scope

This report covers evidence summarized in the provided input across **smart grids, microgrids, distributed energy resources, PV systems, smart meters, inverter control, and federated utility learning pipelines**, with most studies falling in the **2019-2025** period and emphasizing simulations, prototypes, comparative ML studies, and selected embedded deployments [4] [13] [17] [33].

Assumptions & Limitations

- The supplied evidence gives **citation IDs and findings summaries**, but not a complete bibliographic listing for every paper, so this report identifies relevant studies by topic and citation ID rather than full reference metadata [meta direct_answer/unable_to_answer reflected in 1-3,2-2-1-4].
- Grid-specific TinyML evidence is **stronger for conceptual motivation and prototype feasibility than for standardized deployment benchmarking**, especially for flash/RAM/energy/latency trade-offs on actual utility assets [44] [45] [46] [47].
- Much of the literature remains **simulation- or testbed-based rather than field-validated in persistent utility operations** [14] [15] [20] [48] [49].
  Confidence level: High - these limitations are explicitly repeated across the supplied evidence [14] [44] [45].

Suggested Further Research

- Build **grid-specific TinyML benchmark suites** with accuracy, latency, energy, flash, and SRAM reported on the same tasks across device classes [31] [45].
- Run more **field trials on feeders, substations, microgrids, and DER fleets** rather than stopping at simulation or prototype stages [14] [20].
- Report **communication-outage benefits quantitatively** in forecasting and control papers, not just accuracy [22] [44].
- Evaluate **compressed fault models** for transformers, batteries, PV inverters, and smart meters using precision/recall/F1 plus detection delay [14] [50] [51].
- Test **secure federated utility pipelines** under non-IID data, lossy links, and adversarial participants [35] [36] [37] [38].

Full Report

Introduction and Scope

TinyML and edge AI matter for grid management because many power-system decisions are **time-sensitive, geographically distributed, and communication-dependent**. Moving inference closer to smart meters, inverter controllers, substations, or microgrid nodes can reduce communication delay, lower cloud dependence, and improve resilience during connectivity problems [2] [9] [42]. Across the supplied studies, edge intelligence is most justified not by novelty alone but by **local responsiveness** in forecasting, fault monitoring, and control [2] [10] [41] [42].

The evidence base is broad but uneven. Forecasting papers often report strong predictive performance yet omit embedded latency or memory benchmarks [44] [52]. Fault-monitoring papers provide more hardware detail, including Raspberry Pi and ESP32-class deployment figures [11] [12] [13]. Control papers most clearly show operational value, especially in microgrids where AI-assisted droop and voltage regulation beat conventional baselines [17] [19] [20].
Confidence level: High - multiple independent themes in the supplied synthesis align on this point [2] [10].

Foundational Context

What “TinyML” and “edge AI” mean in this literature

In the provided studies, **edge AI** refers to inference or control running near the physical process—on embedded boards, smart meters, inverter controllers, or edge servers—rather than only in centralized cloud systems [53] [54] [55]. **TinyML** is the more constrained end of this spectrum: ML deployed under severe limits in memory, clock speed, energy, and storage, often requiring quantization, pruning, or lightweight architectures [1] [26] [27] [28].

Why this fits grid management

Grid-management tasks are often split into three categories:

1. **Forecasting** for demand, PV output, or irradiance to support dispatch and balancing [1] [3] [4].
2. **Detection/diagnosis** for anomalies, attacks, and maintenance issues requiring fast local alarms [9] [10] [11].
3. **Control** for voltage, frequency, and DER coordination where communication delay can degrade stability [2] [17] [25].

A recurring systems pattern is **local inference for immediate action, centralized or federated retraining for coordination** [22] [24] [39].
Confidence level: High - directly supported by architecture and deployment findings [22] [24] [39] [40].

Edge Forecasting for Load and Renewable Generation

Short-term load forecasting

The forecasting literature does not support a single universally best model. Instead, performance is task- and dataset-specific. In one 2025 Greek distribution-system study, **LightGBM** outperformed CNN, LSTM, GRU, and CNN-LSTM with **MAE 69.12 MW, RMSE 101.67 MW, and MAPE 1.20%** [4]. A six-country comparison likewise found **LightGBM** most robust overall, with **XGBoost** also strong and LSTM/RNN weaker on some datasets [6]. By contrast, other studies favored deep sequence models: **CNN-LSTM** gave the best test RMSE and MAE in the Central Java-DIY system [5], while a separate comparison found **LSTM** best by RMSE and R2, with CNN-LSTM close behind and ARIMA poor [7]. Another study reported that **LSTM** handled long temporal dependencies better, while **XGBoost** remained faster and better for some feature interactions [8].

Taken together, these findings suggest that utilities should not assume "deep learning wins by default"; **tree-based models can be highly competitive or superior when features are structured and efficiency matters**, while compact sequence models remain useful where temporal dynamics dominate [4] [6] [7] [8].
Confidence level: High - several comparative studies explicitly report mixed winners rather than one dominant class [4] [5] [6] [8].

Renewable generation and irradiance forecasting at the edge

The clearest TinyML-aligned forecasting paper in the set is the **Tiny Transformer** study for on-site GHI forecasting, which reports better size, inference time, and accuracy than RNN and ANN baselines [1]. A separate microgrid study showed that **solar inverter power forecasting and droop control were feasible on an ARM-based smart-meter board**, supporting the case for near-device forecasting in distributed energy systems [2]. There is also evidence that simple, parsimonious modeling can be valuable when transmission from remote PV sites is bandwidth-limited [56].

However, many renewable forecasting studies still emphasize predictive accuracy without reporting embedded-board latency, memory footprint, or energy use [52] [57] [58] [59] [60]. This is a major practical gap because forecasting is often presented as "edge-ready" without the deployment metrics needed for hardware selection [44].
Confidence level: Medium-High - the positive forecasting evidence is clear, but hardware-grounded validation is incomplete [1] [2] [44].

Hybrid and collaborative forecasting architectures

Hybrid designs appear to be the most realistic forecasting architecture. An edge-cloud collaborative study using **SE-ResLSTMNet** reports reduced data-transmission latency and improved deployment efficiency and robustness, though the provided summary gives no exact latency number [22]. A hybrid renewable-energy framework combining LSTM forecasting with task scheduling reports **32.5% lower energy consumption**, **97.8% task completion**, and low latency [21]. Another framework preserved forecasting quality during simulated cloud outages via redundant cloud voting [23].

This points to a practical pattern: **use edge or near-edge forecasting for continuity and local action, while cloud layers provide redundancy, heavier computation, or orchestration** [21] [22] [23].
Confidence level: High - consistent across multiple collaborative-architecture studies [21] [22] [23].

Fault Detection, Anomaly Detection, and Predictive Maintenance

TinyML and edge AI are repeatedly framed as good fits for smart-grid fault analytics because local deployment supports **fast inference and lower communication dependence** [9] [10] [61]. Hardware evidence is stronger here than in forecasting. On Raspberry Pi, TinyML anomaly detection has been reported with **sub-40 ms inference** [11]. ESP32-based deployment can fit in **80 KB RAM** and achieve **sub-16 ms inference** [12]. A Raspberry Pi 4–class estimate reports **13.2-106.6 ms inference**, **3.26-3.40 W power draw**, and **43-362 mJ per inference** [13]. More generally, the literature notes that many TinyML targets operate under **<1 MB flash**, around **1000 KB SRAM**, and low-power processors, making model compression central [62] [63].

For predictive maintenance, the evidence is promising but less asset-specific than a utility may want. Reviews point to PV-oriented TinyML for fault detection and maintenance [14] [15], and one maintenance study reports **100% accuracy** with **19.7 KB RAM** for audio anomaly detection, although not in the main target grid-asset classes named in the query [16]. The supplied evidence explicitly notes the lack of studies in target asset classes that simultaneously report **compressed models plus precision/recall/F1 or detection delay** [14] [16] [50] [51].

Interpretability is increasingly treated as deployment-relevant rather than optional. A power-grid GNN explanation study argues interpretability is important for **trust, accountability, and utility adoption** [64]. SHAP-based frameworks are reported to reveal dominant fault causes and support targeted maintenance decisions [65]. Comparative explanation work also finds SHAP and related methods helpful for operator-facing diagnosis [66] [67].
Confidence level: High on feasibility, Medium on utility-grade maturity - device feasibility is directly evidenced, but asset-specific deployment validation remains thin [11] [12] [14].

Distributed Control and Voltage/Frequency Regulation

This is where edge AI appears most operationally mature. A 2024 study explicitly deployed **forecasting and droop control on an ARM-based smart meter-concentrator instead of a cloud server** [2]. A DRL-based inverter voltage regulator outperformed PID with **mean absolute deviation 0.02 vs 0.06** in islanded mode and **synchronization error below 6 Vrms vs up to 15 Vrms** for PID [17]. A deep-learning droop-coefficient tuner reduced **voltage oscillations from 0.028 pu to 0.004 pu** and **frequency fluctuations from 0.025 pu to 0.007 pu** [19]. Another study reports microgrid balancing within **0.1 to 0.3 s** under enhanced droop control [18], and optimized PI/droop control improved overshoot, settling time, and rise time relative to conventional control [20].

Architecture comparisons further strengthen the case for edge or hybrid control. One hybrid smart-grid simulation reports **58 ms edge-control latency**, **more than 50% lower communication bandwidth**, and **11% higher control accuracy** than cloud-only control [42]. A comparative microgrid study found **decentralized control** most resilient under failures, losses, and delays, with the **highest load-served percentage (64.83%)** and the **lowest demand-limit violation duration (40.61 ms)** [41]. A fog/cloud EMS study reports **up to 98.7% latency reduction** versus cloud-only designs [25].

Taken together, these findings imply that **local AI control is especially valuable where milliseconds matter and communications are imperfect**, which is exactly the case for inverter regulation and microgrid coordination [17] [41] [42].
Confidence level: High - the control literature includes multiple direct performance comparisons against conventional and cloud-heavy baselines [17] [19] [25] [42].

Embedded Hardware Constraints and Performance Trade-offs

The hardware story is clear at a general level: TinyML deployment is bounded by **memory, energy, and latency**, so compression and hardware-aware model selection are necessary [28] [55] [68]. The problem is that most strong benchmarking evidence is **not grid-specific** [45] [46] [47]. General embedded studies show what utilities should care about: Pareto-front comparisons among model-platform pairs [29], ARM Cortex-M energy/resource benchmarking [31], and cross-platform comparisons showing FPGA and ASIC options more energy-efficient than GPU-based inference for some workloads [30]. Platform-aware search also shows large efficiency variation depending on model-device pairing, with reported models ranging down to **0.16 mJ per inference** [69].

For grid practitioners, the implication is not that one chip or one model is best, but that **deployment success depends on co-design**: choose the model, compression strategy, and hardware together rather than sequentially [28] [29] [70] [71].
Confidence level: Medium-High - general TinyML evidence is strong, but direct transfer to specific grid workloads remains partly inferential [29] [31].

Deployment Architectures and Learning Pipelines

Federated learning is being explored for grid and energy applications because it keeps data local while allowing distributed model improvement [72] [73] [74]. In the supplied studies, FL improved anomaly detection in distributed energy systems by **15% over centralized approaches** and reached **100% threat detection after seven rounds** [32]. A federated microgrid energy-management framework improved **dispatch efficiency by 18.7%** over traditional centralized optimization [33]. Federated PV prediction also showed useful privacy-accuracy trade-offs relative to centralized edge-based prediction [34].

But FL adds communication and security complexity. Hierarchical and semi-asynchronous variants are proposed to reduce uplink congestion and avoid waiting for slow participants [39] [40]. Clustered or hierarchical FL is repeatedly positioned as more robust under non-IID data and bandwidth constraints [75] [76]. At the same time, the literature stresses vulnerabilities to **poisoning, backdoors, model inversion, membership inference, and Byzantine behavior**, motivating robust aggregation, differential privacy, and secure aggregation [37] [38] [77] [78].

This suggests that utility-scale learning pipelines should be designed as **communication-aware and adversary-aware from the start**, not retrofitted later [38] [79] [80] [81].
Confidence level: High - both benefits and risks are directly and repeatedly documented [32] [33] [37] [39].

Field Validation, Testbeds, and Practical Deployment Lessons

The biggest cross-cutting lesson is that the field is **ahead in concept and prototypes, behind in persistent field deployment** [14] [15] [48] [49]. Microgrid control studies often report strong simulation gains [19] [20], and embedded feasibility has been shown on selected devices [11] [12]. FL testbeds exist for variable wireless conditions [49], and some edge deployments report encouraging latency reductions [25] [42]. Still, the literature repeatedly points to unresolved issues: non-IID data, unreliable links, compute exhaustion, interoperability, and cybersecurity [35] [44] [82] [83].

The most defensible practical conclusion is that **TinyML and edge AI are ready for selective deployment in narrow, latency-sensitive grid tasks**, especially monitoring and local control, but **not yet fully standardized for broad utility rollout across all grid-management functions** [10] [24] [45].
Confidence level: Medium-High - supported by consistent prototype evidence and repeated statements about deployment gaps [14] [25] [45].

Practical Reading List from the Supplied Evidence

If you want a focused set of papers/topics to start with, prioritize these:

1. **Tiny Transformer for on-site GHI forecasting** - best direct TinyML-style forecasting example [1].
2. **ARM smart-meter-concentrator for microgrid forecasting + droop control** - strong edge deployment/control relevance [2].

3. **Comparative short-term load forecasting studies** - for model selection trade-offs: LightGBM vs LSTM/CNN-LSTM/GRU/XGBoost [4] [5] [6] [7] [8].
4. **Raspberry Pi / ESP32 fault-anomaly deployments** - best hardware-feasibility examples [11] [12] [13].
5. **DRL and ML-based microgrid regulators** - strongest evidence of operational control benefit [17] [19] [20].
6. **Hybrid edge-cloud microgrid control comparisons** - best for latency/bandwidth/resilience evidence [25] [41] [42] [43].
7. **Federated learning for anomaly detection, PV prediction, and microgrid EMS** - best for privacy-preserving deployment pipelines [32] [33] [34].

Overall Conclusion

The supplied research supports a clear conclusion: **TinyML and edge AI can improve grid management most convincingly when they are used for local forecasting, real-time anomaly detection, and microgrid voltage/frequency control under latency or connectivity constraints** [2] [10] [42]. Reported benefits include improved forecasting or balancing outcomes, lower communication dependence, faster local response, and better control behavior than cloud-only or conventional baselines in several studies [17] [19] [25] [42] [84]. However, the literature still lacks enough **grid-specific embedded benchmarks and field trials** to make universal deployment claims [14] [44] [45].

Bottom line: for current grid practice, the most evidence-backed strategy is **edge-first for fast monitoring and control, hybrid/federated support for coordination and retraining** [22] [24] [39]. **Confidence level: High** - this conclusion is supported across multiple themes and study types, despite limited field-scale validation [2] [10] [25] [33].

References

1. Hong Vu, T.T., Nguyen, Q.D., Phan, A.T., (...), Bui, V.C. (2025) Global Horizontal Irradiance Prediction Using Interpretable Tiny Machine Learning. *EEE-AM 2025 - IEEE 2nd Asia Meeting on Environment and Electrical Engineering*. https://doi.org/10.1109/EEE-AM66675.2025.11473413
2. Xu, L., Huang, J., Yang, S., Zhu, Y. (2024) Edge Computing for Microgrid via MATLAB Embedded Coder and Low-Cost Smart Meters. *2024 8th International Conference on Smart Grid and Smart Cities, ICSGSC 2024*. https://doi.org/10.1109/ICSGSC62639.2024.10813710
3. Kabir, M.A., Fattah, S. (2024) Machine Learning based Load Forecasting using Efficient Feature Extraction Scheme in Smart Grid. *2024 IEEE 1st International Conference on Green Industrial Electronics and Sustainable Technologies, GIEST 2024*. https://doi.org/10.1109/GIEST62955.2024.10959795
4. Shiblee, M.F.H., Koukaras, P. (2025) Short-Term Load Forecasting in the Greek Power Distribution System: A Comparative Study of Gradient Boosting and Deep Learning Models. *Energies*. https://doi.org/10.3390/en18195060

5. Kharisudin, I., Arissinta, I.O., Aulia, S.A., (...), Wijaya, G.K. (2025) COMPARATIVE STUDY OF LSTM-BASED MODELS WITH HYPERPARAMETER OPTIMIZATION FOR SHORT-TERM ELECTRICITY LOAD FORECASTING. *Barekeng*. https://doi.org/10.30598/barekengvol20no1pp0105-0122
6. Musbah, H., Elsaraiti, M. (2025) Forecasting load consumption: a comprehensive evaluation of deep learning and machine learning techniques. *Electric Power Systems Research*. https://doi.org/10.1016/j.epsr.2025.111834
7. Pandiyan, P., Gurudeep, A.R., Gowtham, M., (...), Bala Subramanian, S. (2025) Electrical Load Demand Forecasting Using Machine Learning Algorithm. *Proceedings of 2025 2nd International Conference on Cognitive Robotics and Intelligent Systems, ICC - ROBINS 2025*. https://doi.org/10.1109/ICC-ROBINS64345.2025.11086163
8. Vijaisrinivaas, A.M.C., Prasad, H., Vigneshwaran, P., Sudhakar, T.D. (2025) Energy Demand Forecasting in an Integrated Power System Using Machine Learning Techniques. *Proceedings - 1st International Conference on Frontier Technologies and Solutions, ICFTS 2025*. https://doi.org/10.1109/ICFTS62006.2025.11031974
9. Trivedi, D., Bhatia, S., Bhatt, S., (...), Tanwar, S. (2025) DeFeAt: Explanable TinyML-based Framework for Anomaly Detection in Nano-Sensor Traffic in 6G-Enabled Smart Grid. *2025 5th Intelligent Cybersecurity Conference, ICSC 2025*. https://doi.org/10.1109/ICSC65596.2025.11140465
10. Janani, S., Sumalatha, V. (2025) Edge-Driven Data Acquisition and Intelligence for Real-Time Energy Management in Smart Grids. *Proceedings of 6th International Conference on Intelligent Communication Technologies and Virtual Mobile Networks, ICICV 2025*. https://doi.org/10.1109/ICICV64824.2025.11085555
11. Nasir, N., Lutui, R., Sarkar, N.I., Vaipulu, T. (2026) Adaptive machine learning based cyber threat intelligence in industrial control systems: A systematic literature review. *International Journal of Critical Infrastructure Protection*. https://doi.org/10.1016/j.ijcip.2026.100874
12. Dedhia, P., Raskar, A., Khodke, S., (...), Pal, A. (2026) Light weight computation offloading approach for edge devices. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2026.111978
13. Hyosun Yoo, H., Sarvi, M., Bagloee, S. (2026) A lightweight hybrid machine and deep learning framework via contrastive learning for audio-based multiclass vehicle counting. *Transportation Research Part C: Emerging Technologies*. https://doi.org/10.1016/j.trc.2026.105631
14. Mellit, A., Zennaro, M. (2025) Recent Advances in the Application of TinyML and Edge Devices in Photovoltaic Systems: Bridging the Gap Between Laboratory Research and Industry. *Proceedings of 2025 International Conference on Artificial Intelligence, Embedded Systems, and Renewable Energy, AIESRE 2025*. https://doi.org/10.1109/AIESRE67541.2025.11470278
15. Jayalakshmi, G., Jayalakshmi, C., Ramesh, M. (2026) Predictive Maintenance Using Tiny Machine Learning: A Revolutionary Approach to Proactive Equipment Maintenance. *Tiny Machine Learning: Fundamentals, Applications, and Security*. https://doi.org/10.1002/9781394347124.ch7

16. Putra, P.D. (2024) Implementation of Audio Classification Using Tiny Machine Learning for Predictive Maintenance in Muara Karang Power Plant. *ICT-PEP 2024 - International Conference on Technology and Policy in Energy and Electric Power: Resilient Power Systems: Navigating the Clean Energy Transition, Proceedings*. https://doi.org/10.1109/ICT-PEP63827.2024.10733513

17. Dinata, N.F.P., Jambak, M.I., Ramli, M.A.M., Bakar Sidik, M.A.B. (2024) Utilizing Deep Reinforcement Learning for Enhanced Microgrid Voltage Regulation under Fluctuating Load Conditions. *ICECOS 2024 - 4th International Conference on Electrical Engineering and Computer Science, Proceeding*. https://doi.org/10.1109/ICECOS63900.2024.10791118

18. Islam, M.A., Rashid Mahin, M.S.R., Kabir, S., Shufian, A. (2024) Mitigating System Imbalance Using Enhanced Droop Control in Decentralized Microgrids. *2024 IEEE 3rd International Conference on Control, Instrumentation, Energy and Communication, CIEC 2024 - Proceedings*. https://doi.org/10.1109/CIEC59440.2024.10468255

19. Khosravi, N., Abdolmohammadi, H.R. (2025) A hierarchical deep learning-based recurrent convolutional neural network for robust voltage and frequency operation management in microgrids. *Applied Soft Computing*. https://doi.org/10.1016/j.asoc.2024.112645

20. Majeed Kareem, R., Al-Nussairi, M.K., Bayindir, R. (2025) Optimal Performance of PI Controller for AC Microgrid Based on Metaheuristic Optimization Algorithms. *IEEE Access*. https://doi.org/10.1109/ACCESS.2025.3546321

21. Amir Latif, R.M.A., Obaidat, M.S., Ullah, F., Mahmood, K. (2026) AI-Driven Energy-Efficient Load Balancing in Hybrid Edge-Cloud Architectures for Renewable Energy Networks. *IEEE Networking Letters*. https://doi.org/10.1109/LNET.2025.3596126

22. Fang, L., Xu, H., Yang, N., Ma, J. (2025) Smart Energy Data Forecasting and Edge Computing Optimization Based on SE-ResLSTMNet. *2025 5th International Conference on Electrical Engineering and Computer Technology, IC2ECT 2025*. https://doi.org/10.1109/IC2ECT66838.2025.11290952

23. Hassanpouri Baesmat, K., Regentova, E.E., Baghzouz, Y. (2025) Cloud-Enabled Hybrid, Accurate and Robust Short-Term Electric Load Forecasting Framework for Smart Residential Buildings: Evaluation of Aggregate vs. Appliance-Level Forecasting. *Smart Cities*. https://doi.org/10.3390/smartcities8060199

24. Wiafe, E., Ofori, C., Boateng, M., (...), Dwomoh, L. (2026) Centralized and Decentralized Energy Management Controls for Microgrids: Architectures and Open Challenges. *2026 International Conference on Cybersecurity, Digital Forensics, and AI Applications, ICCSDFAI 2026*. https://doi.org/10.1109/ICCSDFAI70505.2026.11647894

25. Wakankar, R., Simhachalam, S., Ganesan, R., Velmurugan, T. (2024) Improved Efficiency Of Fog And Cloud Computing Paradigms. *2024 2nd International Conference on Advancements and Key Challenges in Green Energy and Computing, AKGEC 2024*. https://doi.org/10.1109/AKGEC62572.2024.10869278

26. Joshua, R.G., Gbadega, P.A., Imoize, A.L., Tofade, S.O. (2026) TinyML for Anomaly Detection. *Tiny Machine Learning: Design Principles and Applications*. https://doi.org/10.1002/9781394294572.ch3
27. Hizem, M., Bousbia, L., Ben Dhiab, Y., (...), Bouallegue, R. (2025) Reliable ECG Anomaly Detection on Edge Devices for Internet of Medical Things Applications. *Sensors*. https://doi.org/10.3390/s25082496
28. Lamaakal, I., Yahyati, C., Yassine, Y., (...), Ouahbi, I. (2026) TinyML: Small Models Making a Big Impact. *Springer Proceedings in Physics*. https://doi.org/10.1007/978-3-032-19038-3_23
29. Rojas-Carrasco, P., Guinaldo, M. (2026) Multi-Objective Evaluation of Lightweight AI Models on Low-Cost Edge Devices Using Pareto Fronts. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app16083679
30. Ning, Z., Vandersteegen, M., van Beeck, K., (...), Vandewalle, P. (2024) POWER CONSUMPTION BENCHMARK FOR EMBEDDED AI INFERENCE. *Proceedings of the International Conferences on Applied Computing and WWW/Internet 2024*. https://www.scopus.com/pages/publications/85214130241
31. Jain, P., Kasper, M., Köber, G., (...), Seuss, D. (2026) Pareto Optimal Benchmarking of AI Models on ARM Cortex Processors for Sustainable Embedded Systems. *Pervasive Intelligence – From Architectures to Sustainable Edge AI Systems-of-Systems*. https://www.scopus.com/pages/publications/105044797303
32. Mishra, V. (2025) Federated Learning for Enhancing Cybersecurity Resilience in Distributed Energy Systems. *Society of Petroleum Engineers - ADIPEC 2025*. https://doi.org/10.2118/229330-MS
33. Mahmoudi Rashid, S.M., Ghiasi, A.R., Ghavifekr, A.A. (2025) Federated Learning-Based Energy Management Framework for Decentralized Microgrids. *2025 10th International Conference on Technology and Energy Management, ICTEM 2025*. https://doi.org/10.1109/ICTEM66196.2025.11063632
34. Mehran, N., Dorfinger, P., Leschke, N., Pallas, F. (2025) CF-PV: Centralized vs. Federated Edge-based Prediction Models for PV Energy Production. *IEEE PES Innovative Smart Grid Technologies Conference Europe*. https://doi.org/10.1109/ISGTEurope64741.2025.11305643
35. Huang, Wuwei, Yang, Yang, Chen, Mingzhe, (...), Poor, H. Vincent (2021) *Entropy*. https://doi.org/10.3390/e23111413
36. https://www.sciencedirect.com/science/article/pii/S1573754325001721
37. Ye, T., Chen, C., Wang, Y., (...), Gao, M. (2024) BapFL: You can Backdoor Personalized Federated Learning. *ACM Transactions on Knowledge Discovery from Data*. https://doi.org/10.1145/3649316
38. Haque, R.U., Touhidul Hasan, A.S.M., Al-Hababi, M.A.M., (...), Xu, D. (2024) SSI–FL: Self-sovereign identity based privacy-preserving federated learning. *Journal of Parallel and Distributed Computing*. https://doi.org/10.1016/j.jpdc.2024.104907
39. https://www.sciencedirect.com/science/article/pii/S1936645026000884
40. https://www.sciencedirect.com/science/article/pii/S1573754325006955

41. Vasu, S., Kumar, P.R., Jasmin, E.A., Mini, V. (2026) Sustainability-Oriented Priority-Based Load Management Control Architectures for Demand-Constrained Grid-Tied PV–Battery AC Microgrids Using MAS: A Comparative Evaluation. *Sustainability (Switzerland)*. https://doi.org/10.3390/su18168508

42. Akila, P., Gottipalli, D., Saheb, S.N., (...), Kumar Sahoo, D.K. (2025) Design and Simulation of a Hybrid Smart Grid Energy System Using Edge Computing and IoT-Based Control Architectures. *13th IEEE International Conference on Smart Grid, icSmartGrid 2025*. https://doi.org/10.1109/ICSMARTGRID66138.2025.11071723

43. Liya, B.S., Harish Kumar, E., Morshedur Hassan, M.M., (...), Anand, D. (2026) Energy efficient cyber-physical control of renewable microgrids using edge-AI enabled IoT and secure blockchain coordination. *Scientific Reports*. https://doi.org/10.1038/s41598-026-50169-y

44. Wang, Y., Zhao, X., Wang, K., (...), Li, P. (2024) A lightweight method of integrated local load forecasting and control of edge computing in active distribution networks. *iScience*. https://doi.org/10.1016/j.isci.2024.110271

45. Fraga-Lamas, Paula, Lopes, Sérgio Ivan, Fernández-Caramés, Tiago M. (2021) Green IoT and Edge AI as Key Technological Enablers for a Sustainable Digital Transition towards a Smart Circular Economy: An Industry 5.0 Use Case. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s21175745

46. Ray, P.P. (2022) A review on TinyML: State-of-the-art and prospects. *Journal of King Saud University - Computer and Information Sciences*. https://doi.org/10.1016/j.jksuci.2021.11.019

47. Mnif, M., Sahnoun, S., Ben Saad, Y., (...), Kanoun, O. (2024) Combinative model compression approach for enhancing 1D CNN efficiency for EIT-based Hand Gesture Recognition on IoT edge devices. *Internet of Things (The Netherlands)*. https://doi.org/10.1016/j.iot.2024.101403

48. Kholidy, H.A., Kamaludeen, R. (2022) An Innovative Hashgraph-based Federated Learning Approach for Multi Domain 5G Network Protection. *Proceedings - 2022 IEEE Future Networks World Forum, FNWF 2022*. https://doi.org/10.1109/FNWF55208.2022.00033

49. van Rensburg, L.J., Zhao, L. (2025) Towards a Resilient Federated Edge Intelligence: A Testbed for Design, Analysis, and Validation of Federated Learning. *2025 IEEE Opportunity Research Scholars Symposium, ORSS 2025*. https://doi.org/10.1109/ORSS66051.2025.11121622

50. Klein, L., Zmij, P., Krömer, P. (2023) Partial Discharge Detection by Edge Computing. *IEEE Access*. https://doi.org/10.1109/ACCESS.2023.3268763

51. Hamad, M.M., Sheikholeslami, A. (2024) Comparative Analysis of Machine Learning Algorithms for Fault Detection and Localization in Photovoltaic Systems. *2024 3rd International Conference on Smart Grids and Energy Systems, SGES 2024*. https://doi.org/10.1109/SGES63808.2024.10824064

52. Ma, W., Zhou, H., Wu, J., (...), Li, D. (2025) Optimizing spatiotemporal prediction accuracy of regional solar irradiance through multi-angle cloud layer 3D reconstruction. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2025.119814

53. Pau, D.P., Tognocchi, S. (2026) Reviewing progresses on In-Sensor AI Computing. *Microprocessors and Microsystems*. https://doi.org/10.1016/j.micpro.2025.105156
54. Terziyska, M., Ilieva, I., Terziyski, Z., Komitov, N. (2026) A Review of the Transition from Industry 4.0 to Industry 5.0: Unlocking the Potential of TinyML in Industrial IoT Systems. *Sci*. https://doi.org/10.3390/sci8010010
55. Trigkas, A., Piromalis, D., Papageorgas, P. (2025) Edge Intelligence in Urban Landscapes: Reviewing TinyML Applications for Connected and Sustainable Smart Cities. *Electronics (Switzerland)*. https://doi.org/10.3390/electronics14142890
56. Hammond, J.E., Lara Orozco, R.A., Baldea, M., Korgel, B.A. (2024) Short-term solar irradiance forecasting under data transmission constraints. *Renewable Energy*. https://doi.org/10.1016/j.renene.2024.121058
57. Li, Q., Zhou, Z., Zhang, H. (2020) Spatio-temporal modeling with enhanced flexibility and robustness of solar irradiance prediction: A chain-structure echo state network approach. *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2020.121151
58. Bastos, B.Q., Cyrino Oliveira, F.L., Milidiú, R.L. (2021) U-Convolutional model for spatio-temporal wind speed forecasting. *International Journal of Forecasting*. https://doi.org/10.1016/j.ijforecast.2020.10.007
59. Al-Ja'Afreh, M.A.A., Amjad, B., Rowe, K., (...), Marquez, J.L.A. (2023) Optimal planning and forecasting of active distribution networks using a multi-stage deep learning based technique. *Energy Reports*. https://doi.org/10.1016/j.egyr.2023.07.014
60. Shan, S., Li, C., Wang, Y., (...), Wei, H. (2024) A deep learning model for multi-modal spatio-temporal irradiance forecast. *Expert Systems with Applications*. https://doi.org/10.1016/j.eswa.2023.122925
61. El Iysaouy, L.E., Moughraoui, F.Z., Azeroual, M. (2026) Artificial intelligence in smart grids: A comprehensive survey. *AIP Conference Proceedings*. https://doi.org/10.1063/12.0044710
62. Ray, P.P. (2022) A review on TinyML: State-of-the-art and prospects. *Journal of King Saud University - Computer and Information Sciences*. https://doi.org/10.1016/j.jksuci.2021.11.019
63. Saeed, Mamoon M., Saeed, Rashid A., Ahmed, Zeinab E. (2024) TinyML for 5G networks. *TinyML for Edge Intelligence in IoT and LPWAN Networks*. https://doi.org/10.1016/B978-0-44-322202-3.00014-2
64. Bosso, R., Chang, C., Zarif, M., Tang, Y. (2025) Explainable Graph Neural Networks for Power Grid Fault Detection. *IEEE Access*. https://doi.org/10.1109/ACCESS.2025.3591604
65. De Santis, E., Ferro, G., Rizzi, A. (2025) A KAN-SHAP Framework for Fault Detection and Analysis in Smart Grids. *Proceedings of the International Joint Conference on Neural Networks*. https://doi.org/10.1109/IJCNN64981.2025.11227922
66. Li, G., Chen, L., Fan, C., (...), Fang, X. (2023) Interpretation and explanation of convolutional neural network-based fault diagnosis model at the feature-level for building energy systems. *Energy and Buildings*. https://doi.org/10.1016/j.enbuild.2023.113326

67. Sapuppo, F., Patané, L., Comi, A., (...), Xibilia, M.G. (2025) Explainability Analysis of V2G Aggregated Capacity Prediction. *2025 International Conference on Control, Automation and Diagnosis, ICCAD 2025*. https://doi.org/10.1109/ICCAD64771.2025.11099476

68. Yahyati, C., Lamaakal, I., El Makkaoui, K., (...), Yassine, Y. (2026) TinyML: Emerging Applications and Future Research Directions. *Tiny Machine Learning Techniques for Constrained Devices*. https://doi.org/10.1201/9781003544449-10

69. Tu, X., Chen, D., Han, K., (...), Wang, H. (2025) PlatformX: An End-to-End Transferable Platform for Energy-Efficient Neural Architecture Search. *SEC 2025 - Proceedings of the 2025 10th ACM/IEEE Symposium on Edge Computing*. https://doi.org/10.1145/3769102.3770622

70. Kulkarni, V., Jujare, V. (2024) TinyML using neural networks for resource-constrained devices. *TinyML for Edge Intelligence in IoT and LPWAN Networks*. https://doi.org/10.1016/B978-0-44-322202-3.00010-5

71. Altayeb, M., Zennaro, M., Manzoni, P. (2026) Edge ML vs. TinyML: A Comparative Analysis with Experimental Results on Power Efficiency and Model Optimization. *Proceedings - IEEE Consumer Communications and Networking Conference, CCNC*. https://doi.org/10.1109/CCNC65079.2026.11366412

72. Janardhanan, H. (2025) Federated Learning in Edge Computing: Advancements, Security Challenges, and Optimization Strategies. *2025 8th International Conference on Circuit, Power and Computing Technologies, ICCPCT 2025*. https://doi.org/10.1109/ICCPCT65132.2025.11176535

73. Imteaj, A., Thakker, U., Wang, S., (...), Amini, M.H. (2022) A Survey on Federated Learning for Resource-Constrained IoT Devices. *IEEE Internet of Things Journal*. https://doi.org/10.1109/JIOT.2021.3095077

74. Yogeshwari, M., Lavanya, S., Deepa, S., Azman, A.B. (2025) Federated Learning for Sustainable Development Using IoT/Edge Computing System. *Applications of Federated Learning in Technological Advancements: Use Cases and Applications*. https://doi.org/10.1201/9781003532323-4

75. Bi, Q., Ge, L., Jiang, M., (...), Lin, W. (2026) HFLMND: Toward robust and efficient hierarchical federated learning via malicious node detection. *Knowledge-Based Systems*. https://doi.org/10.1016/j.knosys.2026.115270

76. Tulaskar, D.P., Sindhu, B., Chakole, N., (...), Bhaiyya, M. (2026) AI and ML empowering 5G and shaping the 6G future: Models, metrics, architectures, and applications. *ICT Express*. https://doi.org/10.1016/j.icte.2025.11.024

77. Bari, B.S., Yelamarthi, K. (2025) Defending Against Adversarial Threats in Federated Learning: A Security Approach with Differential Privacy and Secure Aggregation. *Proceedings - 2025 IEEE International Symposium on Smart Electronic Systems, iSES 2025*. https://doi.org/10.1109/iSES67504.2025.00090

78. Fan, T., Chen, X., Dong, Y., (...), Jing, W. (2024) Lightweight Secure Aggregation for Personalized Federated Learning with Backdoor Resistance. *Proceedings - Annual Computer Security Applications Conference, ACSAC*. https://doi.org/10.1109/ACSAC63791.2024.00071

79. AlTerkawi, L., Dib, F.K. (2025) Federated Learning for Smart Cities: A Thematic Review of Challenges and Approaches. *Future Internet*. https://doi.org/10.3390/fi17120545

80. Saxena, R.K., Khullar, S. (2026) Introduction to Federated Learning and Threat Landscape. *Adversarial AI and Data Poisoning in Federated Learning*. https://doi.org/10.4018/979-8-3373-6224-3.ch001

81. Alhawas, S., Rassam, M.A. (2026) Federated Learning in Edge Computing: Vulnerabilities, Attacks, and Defenses—A Survey. *Sensors*. https://doi.org/10.3390/s26041275

82. Kandula, R., Dandamudi, J.T., David Amar Raj, R.D.A., (...), Prakasha, K.K. (2026) Explainable AI for Smart Grid Intelligence: A Comprehensive Review of Techniques, Applications and Future Directions. *Energy and AI*. https://doi.org/10.1016/j.egyai.2026.100820

83. Mathew, A., Panchami, V. (2024) A Review on Federated Learning with a Focus on Security and Privacy. *RAICS - IEEE Recent Advances in Intelligent Computational Systems*. https://doi.org/10.1109/RAICS61201.2024.10689882

84. Kumar, A., Nandanwar, S.R. (2025) Optimization of Renewable Energy Integration in Smart Grids Using AI-Based Forecasting and Load Balancing Techniques. *2025 9th International Conference on Computing, Communication, Control and Automation, ICCCBEA 2025*. https://doi.org/10.1109/ICCUBEA65967.2025.11284115